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DBMS Assignment - 01

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Q What is Database Management System? Why is it required?

Database Management System is a software which is used to manage the database. It allows users to create, manage, store, retrieve & manipulate data in structured way.
Examples: MySQL, Oracle

It is required to store large amounts of data efficiently, to reduce data redundancy, to maintain data consistency, to provide data security, to provide backup & recovery, to allow multiple users to access data simultaneously.

Q Why is DBMS better than the traditional file system?

Because DBMS reduces data redundancy, ensures data consistency, provide security, support multi-user access & data independence whereas Traditional file system have high data redundancy and limited security, limited multi-user support & No proper recovery system, no data independence and also here data inconsistency possible.

Q What is Data Abstraction? Explain its levels.

Data Abstraction means hiding complex database details and showing only essential information to users.

1. Physical Level (Internal Level):- This is the lowest level of data abstraction. It describes how data is actually stored in DB. At this level, complex details like data blocks, pointers & storage technique are handled by system.
It includes :- Storage structure, Indexing, file-organization & Memory allocation.

2. Logical level (Conceptual level) :→ This is the middle level of abstraction.
It describe what data is stored in dB & relationship among data.
It Includes : Entities, Attributes, Relationship, Constraints.
DB administrators works mainly at this level.

3. View Level (External level) :→ This is highest level of abstraction.
It describe how users see the data. Different users may have diff. views of same database.

Q4. What is the Three-Schema Architecture? Why is it important.

1. View Level {External Schema}
2. Logical Level {Conceptual Schema}
3. Physical Level {Internal Schema}

The three schema architecture is important bcz:-

1. It Provides data Independence. It changes at one Level & do not affect other levels.
2. It improves security. Users can see only the data they allowed to see.
3. It Reduces complexity. It separates the user view from physical storage.
4. It is easy to maintenance.

Q What is Data Independence? Why is it needed?
Data Independence is ability to modify schema at one level without affecting other levels.
Types are :-

1) Physical Data Independence :- Changes only occur in physical storage.
2) Logical Data Independence :- Changes in logical schema doesn't affect external views.

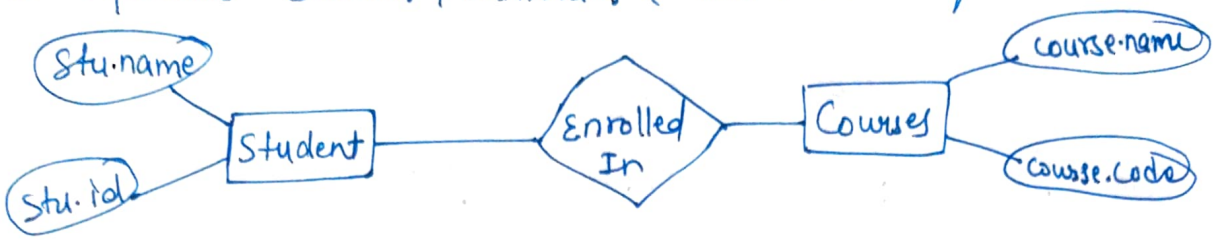
It is needed because to make dB flexible, easy maintenance and reduces application changes.

Q What is ER Model?

ER model stands for an Entity-Relationship model. It is high level data model. This model is used to define the data elements & relationship for a specified system.

It develops a conceptual design for dB. It also develops a very simple & easy to design view of data.

It represents Entities, attributes & the relationship.



Q What are Integrity Constraints? Why are they important?

Integrity constraints in a DBMS are rules that help keep the data in dB accurate, consistent & reliable. They act like a set of guidelines that ensures all the information stored in dB follows specific standards.

Types :- 1) Domain Constraints → Not NULL Constraints, → CHECK constraints

2) Entity Integrity Constraints

3) Key Constraints
   - → Primary Key Constraints
   - → Unique Key Constraints

4) Referential Integrity constraints

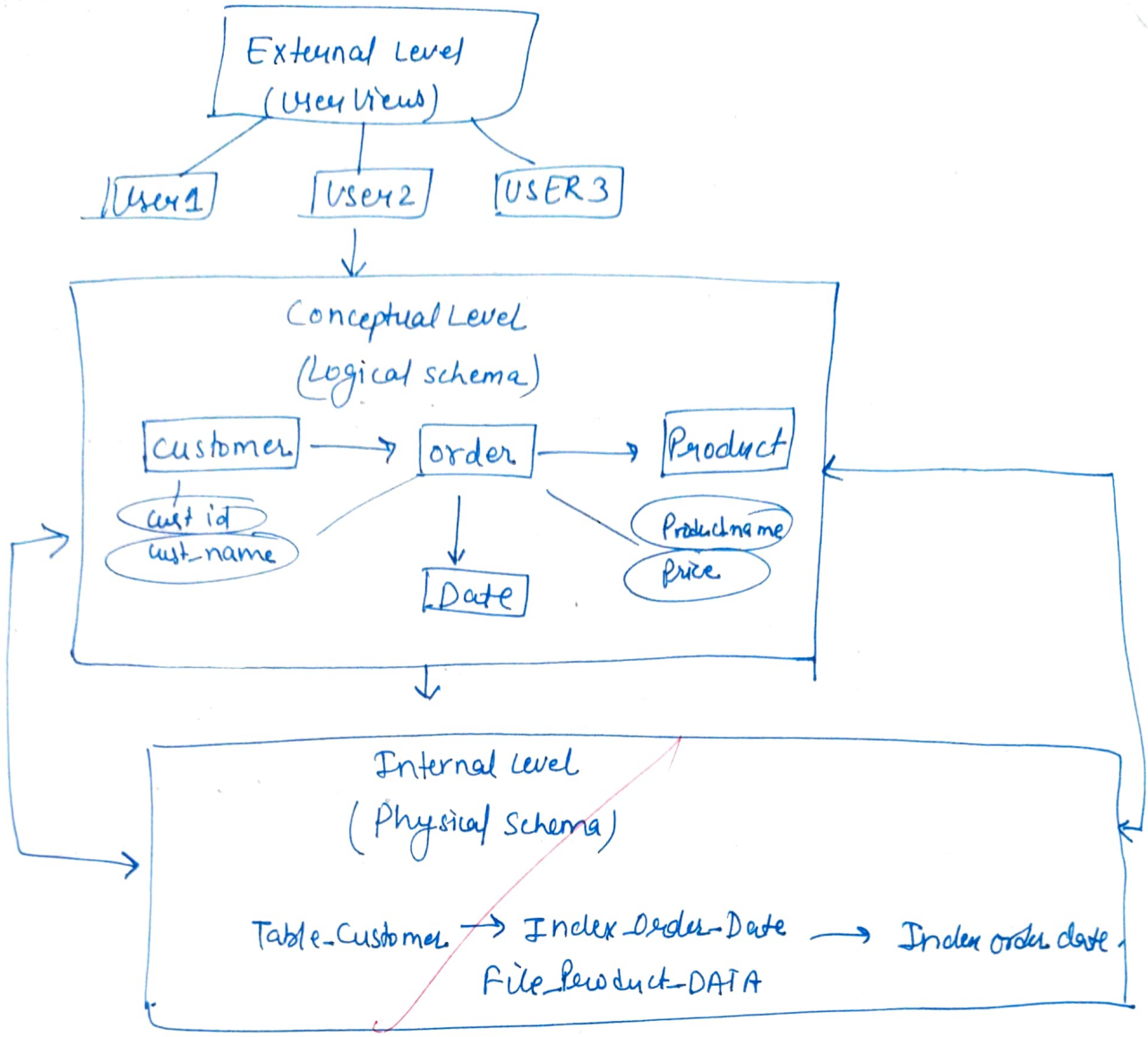
5) Assertion

Importance :-

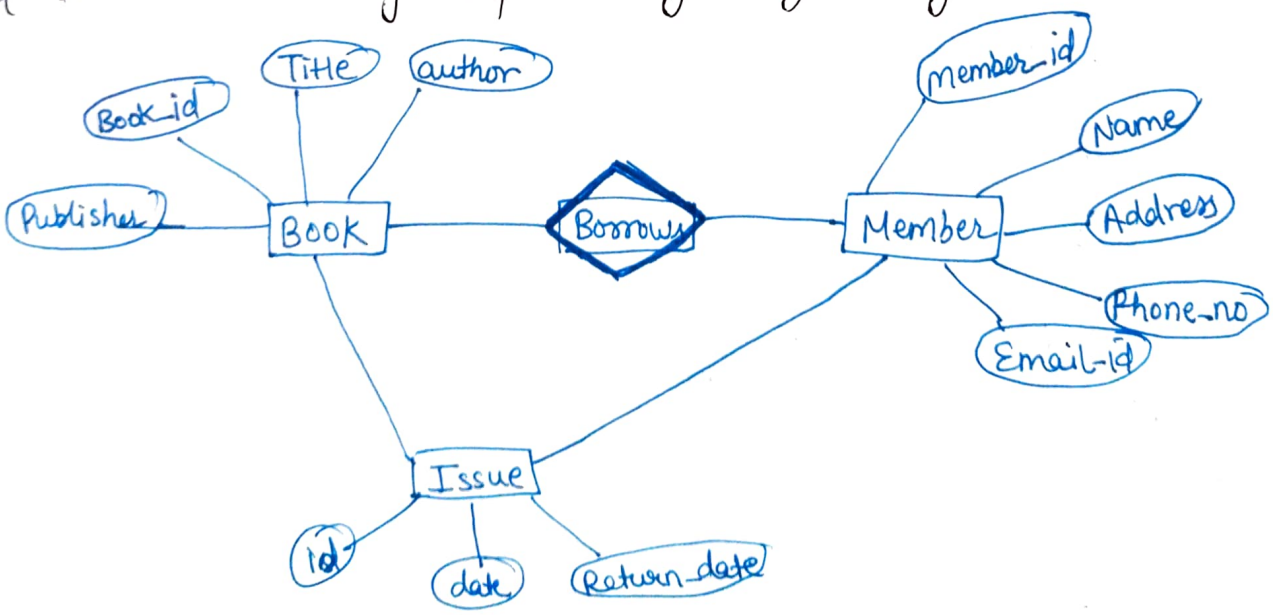
- They prevents invalid data entry and maintains data consistency and also ensures the correct relationship.
- Ensures unique identification of rows.
- Ensures dB is reliable & follows the predefined business rules.
- They make dB easier to manage & maintain by reducing errors.

Section-B

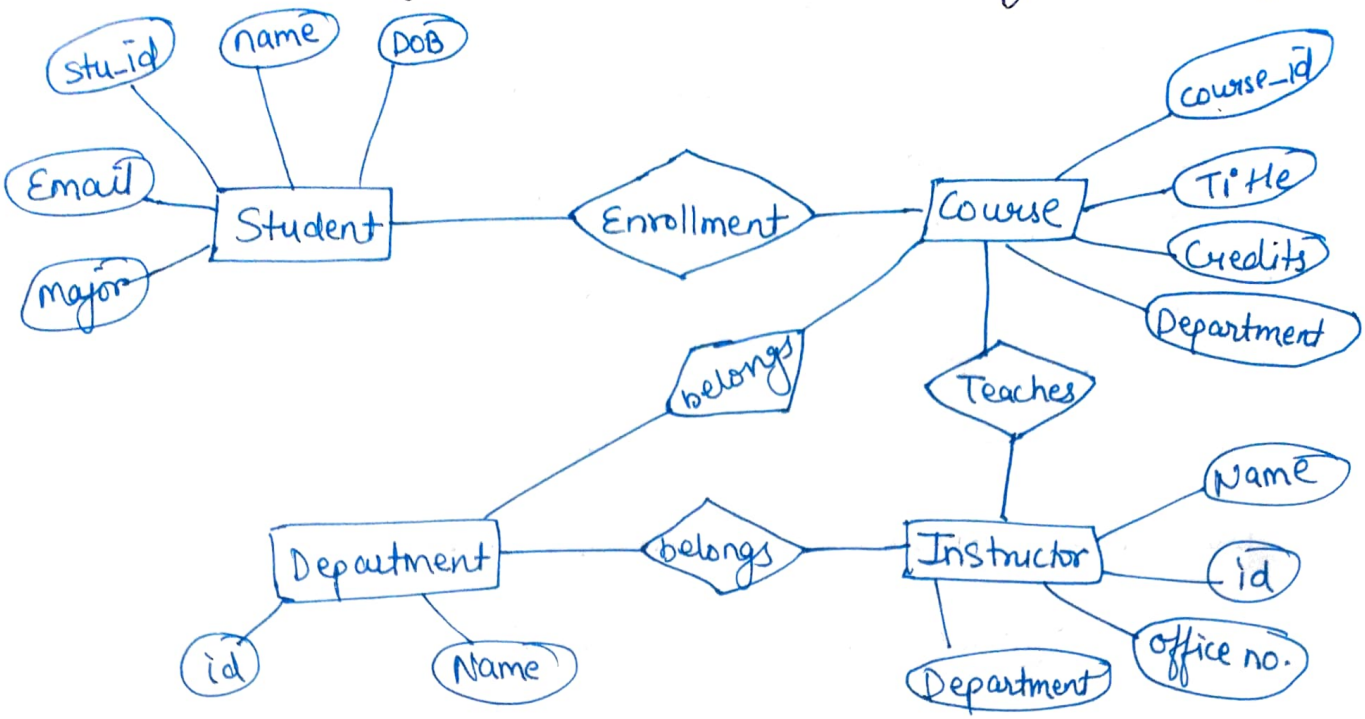
Q Draw the three-schema architecture & explain each level.



Q Draw an ER diagram for Library Management System.



Draw an ER diagram for Student - Course Registration System.



PK = ID FK = Courseid

Section-C

Q. A hospital wants to store patient doctor & appointment data. Identify entities, attributes & relationship.

Entity → Patient → address, pat_id, Name, DOB, Phone, Gender } → attributes

Doctor is Entity & the doctor_id (PK), Name, Speciliazation, License no., Phone no. are attributes

Appointment is entity & id (PK), Date, Time, Status, Room no are attributes

Medical Record is entity & id(PK), Diagnosis, Treatment are attributes

Relationships

Doctor — Schedules — Appointment

Patient — books — Appointment

Doctor — Treats — Patient

Patient — Has — Medical Records

Q. An online shopping platform stores customer, product, & Suggest Keys & Constraints.

1) Primary Keys is the unique identifier for every record. It must be unique & cannot be null
→ Customer_id, Product_id, Order_id

2) Foreign Keys create the links b/w your tables, maintaining Referential Integrity. → Order_Items.

Customer	Email	UNIQUE
Product	Price	CHECK (Price > 0)
	Stock Quantity	DEFAULT 0
Order	Order Date	NOT NULL
order_items	Quantity	CHECK (Quantity >= 1)

Q A University stores student records. Explain how data abstraction helps diff. users.

Data abstraction in university dB ensures that users only interact with the information they need & complex details remain hidden.

1. External Level:- This is what the user sees.

Student View - student can see grades & register for classes but can't see other student's finanical aid & staff salaries.

Professor View - Teachers can put grades & see class rosters, but can't change a student billing status.

2. Conceptual View:- It defines what data exists & how they relate. If the university adds a new "Email" field to all students, this is where it's defined. Bcz of Logical Data Independence, the existing apps don't break, they just recognize the new field.

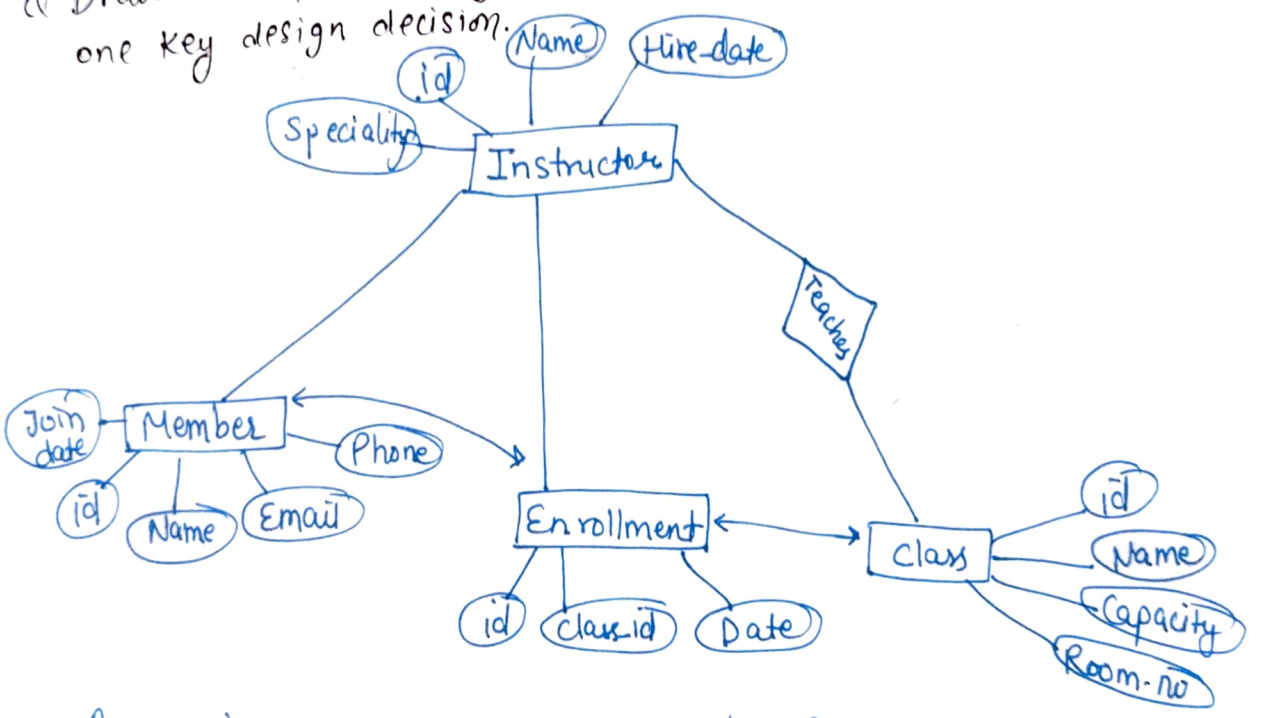
3. Internal Level :- It deals with file paths, indexing for speed & data compression.

The university can switch from an old server to a modern cloud based system without the students or professor even noticing a change in their login screen.

It is important for the user security, simplicity & flexibility.

Section-D

Q Draw a simple ER Diagram for any real life system & explain one key design decision.



A member can enroll in many class & class have many member (N:M).

Q Draw the three-Schema architecture & mention one benefit.

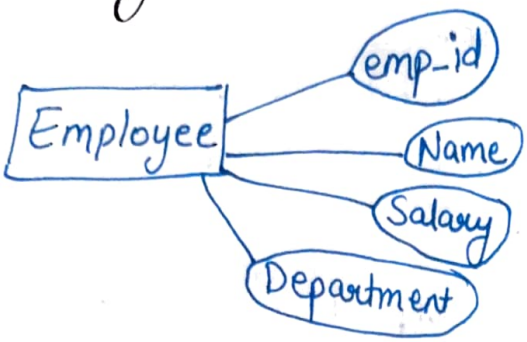
External level
↓
Conceptual level
↓
Internal level.

Data Independence :- Changes in Physical storage do not affect users.

Changes in logical structure do not affect user views.

Makes dB flexible & easy to maintain.

Q. Draw an entity with attributes & identify its Primary Key



Here Employee is Entity and emp_id is Primary Key bcz uniquely identifies each employee and No duplicates allowed and also ensures the entity integrity.